Methods: the federated active-inference stack

This section develops the federated generalized variational inference (FedGVI) core in the discrete-categorical setting and defines the primitives whose recovery limits sec. 6 then states as numbered theorems. Every belief here is a categorical pmf — a non-negative vector summing to one — so the generalized-variational-inference machinery reduces to closed forms that are exactly testable. All mathematics lives in `src/fedference/`; the prose names the module and the identity that pins each claim.

The active-inference community has built a rich apparatus for federated belief sharing: discrete-state-space agents that broadcast posteriors and fuse them into a consensus (Da Costa et al. 2020; Friston et al. 2024), message-passing toolboxes that make exact-Bayes inference scalable (Heins et al. 2022; Bagaev et al. 2023), and collective and multi-agent formulations in which ensembles coordinate by sharing observations and beliefs (Heins et al. 2024; Albarracin et al. 2022; Kaufmann et al. 2021). We accept that apparatus and extend it: the consensus rule the field uses is exact-Bayes and trusting, with no account of what happens when an agent in the ensemble is misspecified or adversarial. Outside active inference, the federated-learning and robust-Bayes literatures address important parts of that question — decentralized aggregation (McMahan et al. 2017), partitioned and federated variational inference (Ashman et al. 2022; Bui et al. 2018), and generalized, robustness-bearing Bayesian updating (Bissiri et al. 2016; Knoblauch et al. 2022; Basu et al. 1998; Zhang and Sabuncu 2018) — but none has been carried into the generative-model-bearing, action-selecting POMDP setting. The methodology below is the bridge: it federates the FedGVI objective (Mildner et al. 2025) per agent inside an active-inference ensemble, proves the standard-Bayes client limits, and tests the project-local zero-robustness log-linear-pool identity. Under the qualified categorical bridge of sec. 5.8, that pool specializes Eq. 7's message-combination term rather than the complete source protocol. fig. 2 illustrates the three-axis architecture and the recovery hierarchy.

A colony of N agents shares a single latent factor $s \in \{1, \dots, n_s\}$ (in the sentinel scenario, the location of a creature on a grid of n_s cells). Each round proceeds in three steps:

Federation protocol: local update, server fusion, broadcast I

1. **Local inference.** Agent n observes o_n and forms a local posterior $q_n(s)$ over the shared factor by a generalized-Bayes update against its own cavity (the colony belief with agent n's previous contribution removed). This is where robustness enters per agent: the update minimizes a loss-plus- divergence objective, and the FedGVI choice of a bounded loss is what carries the source theorem's bounded-influence result under its matching assumptions.
2. **Broadcast.** Agent n broadcasts $q_n(s)$, optionally with a scalar base weight $w_n \geq 0$.
3. **Aggregation.** The server (or, equivalently, each agent acting as its own server) fuses the broadcast beliefs into a consensus. Following sensory attenuation — "agents do not hear themselves" — an agent's heard consensus excludes its own message.

The protocol has two distinct places where robustness can live, and we keep them separate throughout. The **per-agent generalized-Bayes update** in step 1 is FedGVI-faithful at the stated primitive level: its formal bounded-influence claim is conditional on the source theorem's assumptions. The **server-side aggregation rule** in step 3 admits an optional divergence-reweighting heuristic that down-weights agents far from the emerging consensus; this heuristic is a complementary device whose positive formal property is recovery of the naive consensus in its trusting limit, while a scoped proposition rejects one declared separable objective class. sec. 3.3 holds this boundary; no figure, table, or sentence in this work grants the server-side heuristic the per-agent FedGVI guarantee.

Notation for beliefs, losses, and divergences I

The authoritative symbol and API contract is sec. 13. In the main text, $q_n(s)$ denotes agent n's local posterior, $q(s)$ the global consensus, and $q_{-n}(s)$ the cavity after removing the site factor $t_n(s)$. The prior is $\pi_0(s)$, while π is a policy. The POMDP tensors are $A[o, s]$, $B[s', s, u]$, $C[o]$, and $D_0[s]$. The aggregation weights are w_n (raw/base), a_n (raw variational effective), and $\tilde{a}_n$ (normalized influence). The server robustness coefficient is c, the variational entropy weight is λ, the Rényi order is α, the density-power parameter is β, and the robust cross-entropy parameter is q_{loss}. The notation supplement also defines the seed/trial nesting and all statistical quantities used below.

Notation for beliefs, losses, and divergences I

The study is run over a fixed ensemble of 7 agents sharing a factor of 9 locations, with all randomness seeded at 0; the full per-study configuration is tabulated in tbl. 1. As an independent generative-model-free baseline, we also implement FedGVI in a deterministic MLP complement trained with the density-power β-loss — generalized variational inference with a point-mass variational family (sec. 7.6). The remaining methodology subsections develop each primitive in turn: the generalized-Bayes update and its recovery to standard Bayes (sec. 5.3), the divergence family and its KL limit (sec. 5.6) and the robust loss family and its NLL limit (sec. 5.7), the aggregation identity (sec. 5.8), the lift to a belief-sharing round (sec. 5.9), and the paired statistics that earn every "robust beats naive" verdict (sec. 5.27).